

Group 62 Stats 20 Project

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CORRELATION BETWEEN VIOLATION AND GRADE

```
# -----  
# SUMMARY OF THE DATASET  
# -----  
df <- read.csv("newyork_restaurant_inspections.csv")  
head(df)
```

```

##          CAMIS                      DBA          BORO BUILDING
## 1 50179007          STOUT MARKETS INC Manhattan      1328
## 2 50167375          181Exclusive LLC Manhattan      810
## 3 50178660          PAPACHENCHARSIBAO CORP  Queens    61-17
## 4 50111182          THE ALCHEMISTS KITCHEN Manhattan    117
## 5 50075461          POPEYES Manhattan      2300
## 6 50175948 A DAUGHTER AND TWO SON'S LLC  Bronx      1807
##          STREET ZIPCODE          PHONE CUISINE.DESCRPTION
## 1          BROADWAY  10001 7187919245
## 2          WEST 181 STREET  10033 2127950550
## 3          SPRINGFIELD BOULEVARD  11364 3472350424
## 4          CROSBY STREET  10012 9177433747
## 5 ADAM CLAYTON POWELL JR BOULEVARD  NA 6463689765      Chicken
## 6          ARCHER STREET  10460 3477204188
##  INSPECTION.DATE                      ACTION
## 1      01/01/1900
## 2      01/01/1900
## 3      01/01/1900
## 4      01/01/1900
## 5      04/30/2024 Violations were cited in the following area(s).
## 6      01/01/1900
##  VIOLATION.CODE          VIOLATION.DESCRPTION  CRITICAL.FLAG SCORE GRADE
## 1                                Not Applicable  NA
## 2                                Not Applicable  NA
## 3                                Not Applicable  NA
## 4                                Not Applicable  NA
## 5      09B Thawing procedure improper.  Not Critical  22
## 6                                Not Applicable  NA
##  GRADE.DATE RECORD.DATE          INSPECTION.TYPE Latitude
## 1          11/24/2025                      40.75031
## 2          11/24/2025                      40.85103
## 3          11/24/2025                      40.74880
## 4          11/24/2025                      40.72426
## 5          11/24/2025 Cycle Inspection / Initial Inspection  0.00000
## 6          11/24/2025                      40.83754
##  Longitude Community.Board Council.District Census.Tract  BIN  BBL
## 1 -73.98738      105          4      7600 1015867 1008360001
## 2 -73.93876      112         10     26500 1064298 1021770100
## 3 -73.75655      411         23    137700 4163156 4076440009
## 4 -73.99662      102          1     4300 1007930 1005100006
## 5   0.00000      NA          NA      NA      NA      1
## 6 -73.86693      209         18    21800 2028633 2039200024
##  NTA                      Location
## 1 MN17 POINT (-73.987378651873 40.750308585404)
## 2 MN36 POINT (-73.938763711348 40.851030689063)
## 3 QN42 POINT (-73.756547103858 40.748799391976)
## 4 MN24 POINT (-73.9966159052 40.724261421332)
## 5
## 6 BX08 POINT (-73.866927323367 40.837537207937)

```

```
str(df)
```

```
## 'data.frame':    292557 obs. of  27 variables:
## $ CAMIS          : int  50179007 50167375 50178660 50111182 50075461 50175948
50178496 50165833 50150463 50175363 ...
## $ DBA            : chr  "STOUT MARKETS INC" "181Exclusive LLC" "PAPACHENCHARSI
UBAO CORP" "THE ALCHEMISTS KITCHEN" ...
## $ BORO           : chr  "Manhattan" "Manhattan" "Queens" "Manhattan" ...
## $ BUILDING        : chr  "1328" "810" "61-17" "117" ...
## $ STREET          : chr  "BROADWAY" "WEST 181 STREET" "SPRINGFIELD BOULEVARD"
"CROSBY STREET" ...
## $ ZIPCODE         : int  10001 10033 11364 10012 NA 10460 10004 10022 10303 NA
...
## $ PHONE           : chr  "7187919245" "2127950550" "3472350424" "9177433747"
...
## $ CUISINE.DESCRPTION : chr  "" "" "" "" ...
## $ INSPECTION.DATE   : chr  "01/01/1900" "01/01/1900" "01/01/1900" "01/01/1900"
...
## $ ACTION           : chr  "" "" "" "" ...
## $ VIOLATION.CODE     : chr  "" "" "" "" ...
## $ VIOLATION.DESCRPTION: chr  "" "" "" "" ...
## $ CRITICAL.FLAG     : chr  "Not Applicable" "Not Applicable" "Not Applicable" "No
t Applicable" ...
## $ SCORE            : int  NA NA NA NA 22 NA NA NA NA NA ...
## $ GRADE             : chr  "" "" "" "" ...
## $ GRADE.DATE        : chr  "" "" "" "" ...
## $ RECORD.DATE       : chr  "11/24/2025" "11/24/2025" "11/24/2025" "11/24/2025"
...
## $ INSPECTION.TYPE   : chr  "" "" "" "" ...
## $ Latitude          : num  40.8 40.9 40.7 40.7 0 ...
## $ Longitude         : num  -74 -73.9 -73.8 -74 0 ...
## $ Community.Board    : int  105 112 411 102 NA 209 101 105 501 NA ...
## $ Council.District   : int  4 10 23 1 NA 18 1 4 49 NA ...
## $ Census.Tract       : int  7600 26500 137700 4300 NA 21800 900 10000 32300 NA ...
## $ BIN                : int  1015867 1064298 4163156 1007930 NA 2028633 1000812 108
4771 5169467 NA ...
## $ BBL               : num  1.01e+09 1.02e+09 4.08e+09 1.01e+09 1.00 ...
## $ NTA               : chr  "MN17" "MN36" "QN42" "MN24" ...
## $ Location           : chr  "POINT (-73.987378651873 40.750308585404)" "POINT (-7
3.938763711348 40.851030689063)" "POINT (-73.756547103858 40.748799391976)" "POINT (-73.
9966159052 40.724261421332)" ...
```

```
nrow(df)
```

```
## [1] 292557
```

```
ncol(df)
```

```
## [1] 27
```

Hypothesis 1: Worse grades are associated with higher number of violations (Correlation between grade and number of violations)

False Negative: Some A-graded restaurants have violations(especially critical ones)

```
# -----
# KEEP ONLY NEEDED COLUMNS
# -----
# CAMIS          -> Restaurant ID
# DBA            -> Restaurant name
# INSPECTION.DATE -> Date of inspection
# VIOLATION.CODE -> Type of violation
# VIOLATION.DESCRPTION -> Description of violation
# CRITICAL.FLAG  -> Whether violation was critical
# SCORE          -> Numerical inspection score
# GRADE          -> Assigned letter grade (A, B, C)
# ACTION         -> Action taken (e.g., closed, fined, etc.)

df_needed <- df[, c("CAMIS",
                   "DBA",
                   "INSPECTION.DATE",
                   "VIOLATION.CODE",
                   "VIOLATION.DESCRPTION",
                   "CRITICAL.FLAG",
                   "SCORE",
                   "GRADE",
                   "ACTION")]
```

```
# -----
# CLEAN DATA
# -----

# ----- CAMIS -----
#Remove rows with missing ID
df_needed <- df_needed[!is.na(df_needed$CAMIS), ]

# ----- INSEPCTION.DATE -----
#Convert to Date type and remove missing dates
df_needed$INSPECTION.DATE <- as.Date(df_needed$INSPECTION.DATE,
                                     format = "%m/%d/%Y")

#Check for missing values
sum(is.na(df_needed$INSPECTION.DATE))
```

```
## [1] 0
```

```
# ----- VIOLATION.CODE -----
#Check for missing values
sum(is.na(df_needed$VIOLATION.CODE))
```

```
## [1] 0
```

```
# ----- CRITICAL.FLAG -----
df_needed$CRITICAL.FLAG <- trimws(df_needed$CRITICAL.FLAG) #remove spaces
df_needed$CRITICAL.FLAG[df_needed$CRITICAL.FLAG == ""] <- NA
df_needed$CRITICAL.FLAG <- toupper(df_needed$CRITICAL.FLAG) #make everything uppercase
for comparison

table(df_needed$CRITICAL.FLAG, useNA = "ifany")
```

```
##
##          CRITICAL NOT APPLICABLE    NOT CRITICAL
##          154407                8038        130112
```

```
# ----- GRADE -----
#Remove row with missing Grade
df_needed <- df_needed[df_needed$GRADE != "", ]
table(df_needed$GRADE, useNA = "ifany")
```

```
##
##      A      B      C      N      P      Z
## 96869 18272 13108  7415   921  6059
```

```
# -----
# DATA ANALYSIS
# -----

#Create a data frame to count number of violations
df_needed$INSPECTION_ID <- paste(df_needed$CAMIS,
                                df_needed$INSPECTION.DATE,
                                df_needed$GRADE,
                                df_needed$SCORE,
                                sep = "_")

inspection_df <- df_needed %>%
  group_by(INSPECTION_ID, CAMIS, INSPECTION.DATE, GRADE, SCORE) %>%
  summarize(
    num_violations = sum(!is.na(VIOLATION.CODE)),
    num_critical   = sum(CRITICAL.FLAG == "CRITICAL", na.rm = TRUE) # count critical vi
olations
  ) %>%

  ungroup()
```

```
## `summarise()` has grouped output by 'INSPECTION_ID', 'CAMIS',
## 'INSPECTION.DATE', 'GRADE'. You can override using the `.groups` argument.
```

```
# Calculate average number of violations per grade (Only A, B, C)
inspection_df <- inspection_df %>%
  filter(GRADE %in% c("A", "B", "C"))

inspection_df %>%
  group_by(GRADE) %>%
  summarize(
    mean_violations = mean(num_violations),
    n = n()
  )
```

```
## # A tibble: 3 × 3
##   GRADE mean_violations    n
##   <chr>          <dbl> <int>
## 1 A              2.18 44398
## 2 B              4.16  4393
## 3 C              5.74  2285
```

```
# I used AI here to find out which hypothesis test is the best and how to use the function
# Correlation Test
inspection_df <- inspection_df %>%
  filter(GRADE %in% c("A", "B", "C")) %>%
  mutate(
    grade_num = case_when(
      GRADE == "A" ~ 1,
      GRADE == "B" ~ 2,
      GRADE == "C" ~ 3,
      TRUE ~ NA_real_
    )
  )

cor.test(inspection_df$grade_num, inspection_df$num_violations) #AI
```

```
##
## Pearson's product-moment correlation
##
## data: inspection_df$grade_num and inspection_df$num_violations
## t = 202.6, df = 51074, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.6626753 0.6722921
## sample estimates:
##      cor
## 0.6675115
```

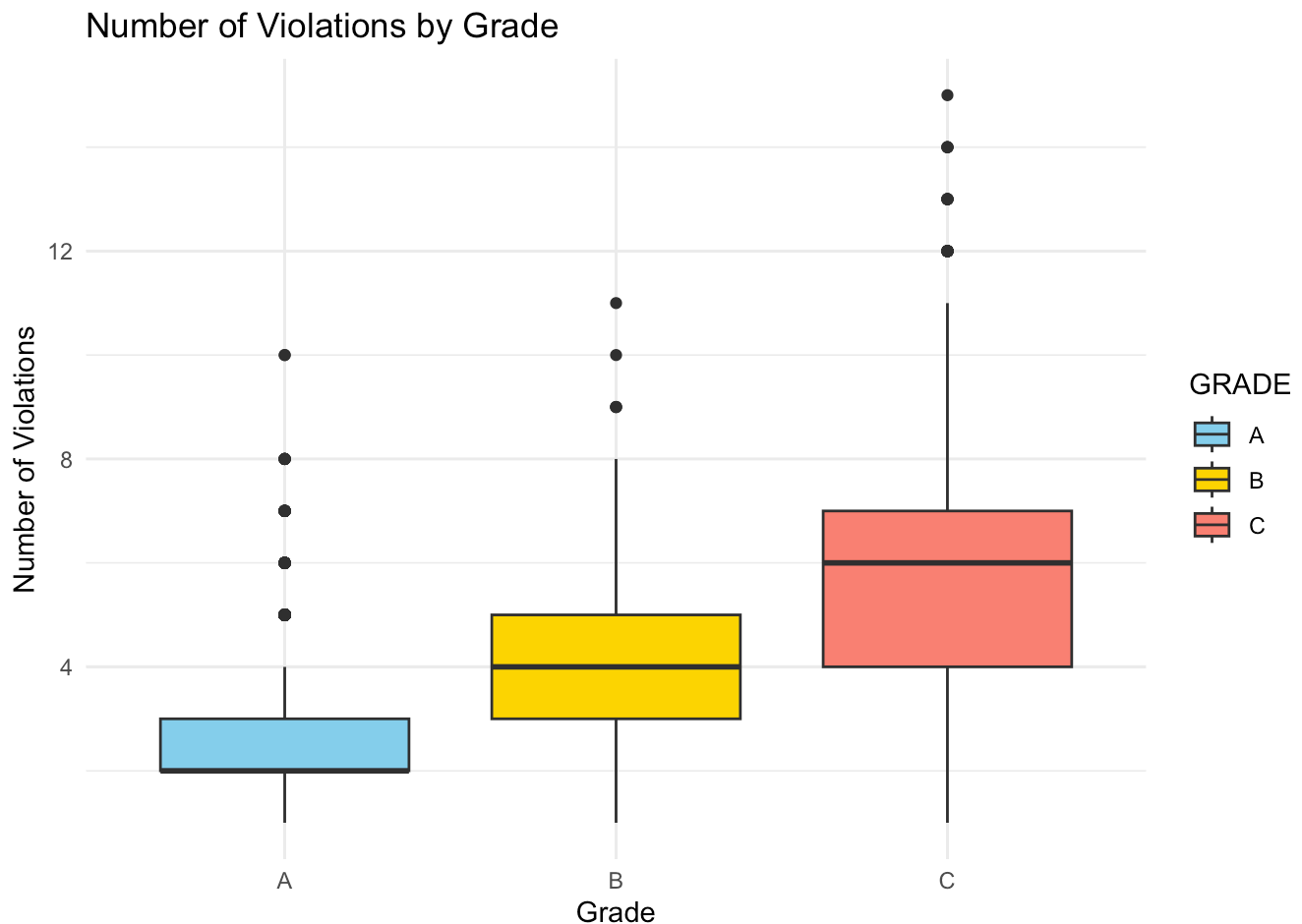
Correlation coefficient: $r = 0.6677886$ There is a moderately strong positive linear relationship between Grade and Number of Violations. As grade worsens (A->B->C :: 1->2->3), the number of violation increases

p-value < 2.2e-16 which is extremely small. Null Hypothesis is rejected. (No linear relationship between Grade and Number of violations)

95% confidence interval: [0.6629520 0.6725695] The true correlation in the population is between 0.663 and 0.73 (far from 0). The relationship is strong and real

The data strongly supports H1: Worse grades are associated with higher number of violations

```
# -----
# PLOT
# -----
# ggplot 2
inspection_df %>%
  filter(GRADE %in% c("A", "B", "C")) %>%
  ggplot(aes(x = GRADE, y = num_violations, fill = GRADE)) +
  geom_boxplot() +
  scale_fill_manual(values = c("skyblue", "gold", "salmon")) +
  labs(
    title = "Number of Violations by Grade",
    x = "Grade",
    y = "Number of Violations"
  ) +
  theme_minimal()
```

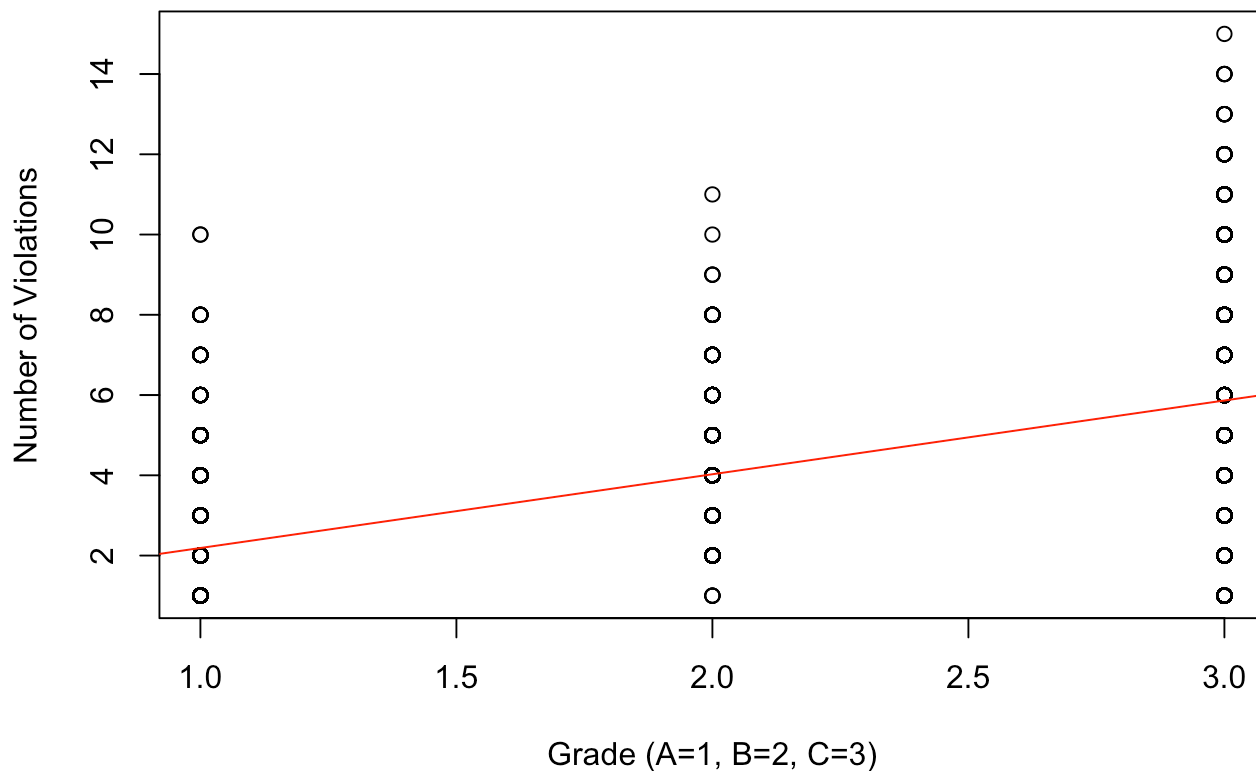


```
ggsave("num_violation_by_grade.png", width = 8, height = 6, dpi = 300)

# Scatter plot to see the trend
plot(inspection_df$grade_num, inspection_df$num_violations,
     xlab = "Grade (A=1, B=2, C=3)",
     ylab = "Number of Violations",
     main = "Correlation Between Grade and Violations")

abline(lm(num_violations ~ grade_num, data = inspection_df), col="red")
```

Correlation Between Grade and Violations



```
ggsave("scatter_corr_grade_violation.png", width=8, height=6, dpi=300)
```

```
# -----
# FALSE NEGATIVES ANALYSIS
# -----
# Total number of A-grade inspections
total_A <- sum(inspection_df$GRADE == "A")

# A-graded inspections with ANY violations
A_any <- sum(inspection_df$GRADE == "A" & inspection_df$num_violations > 0)

# A-graded inspections with CRITICAL violations
A_crit <- sum(inspection_df$GRADE == "A" & inspection_df$num_critical > 0)

# Proportions
prop_any <- A_any / total_A
prop_crit <- A_crit / total_A

total_A
```

```
## [1] 44398
```

```
A_any
```

```
## [1] 44398
```

```
A_crit
```

```
## [1] 36112
```

```
prop_any
```

```
## [1] 1
```

```
prop_crit
```

```
## [1] 0.81337
```

All inspections on A-graded restaurants had at least one documented violations. 81.36% of these inspections had at least one critical violation, meaning a violation that poses a direct risk to public health (e.g., improper food temperatures, evidence of rodents, or contamination hazards).

A grade suggests high food safety, yet the inspection documents violations, including critical ones. Consequently, an A grade does not guarantee the absence of health hazards; rather, it only reflects that the total violation score remained below the numerical threshold for a B or C grade.

PERSISTENCE AND RECIDIVISM

Loading and Cleaning

```
restaurant <- read.csv("newyork_restaurant_inspections.csv")

# Remove columns: Latitude, Longitude, Community.Board, Council.District, Census.Tract,
BIN, BBL, NTA, Location, BUILDING, STREET, ZIPCODE, PHONE, RECORD.DATE
restaurant <- restaurant[, !(names(restaurant) %in% c("Latitude", "Longitude", "Communi
ty.Board", "Council.District", "Census.Tract", "BIN", "BBL", "NTA", "Location", "BUILDIN
G", "STREET", "ZIPCODE", "PHONE", "RECORD.DATE"))]

# Date formatting and filter all rows so that all inspections happened after 1996
restaurant$INSPECTION.DATE <- as.Date(restaurant$INSPECTION.DATE, format = "%m/%d/%Y")
restaurant <- restaurant %>% filter(INSPECTION.DATE >= as.Date("1996-01-01"))

# Remove all NAs for score
restaurant <- restaurant %>% filter(!is.na(SCORE))
```

```
# We filter inspections by date. If different scores in the same inspection date, we wil
l take the highest as the score
restaurant_unique <- restaurant %>%
  group_by(CAMIS, DBA, INSPECTION.DATE) %>%
  summarise(
    SCORE = max(SCORE),
    CRITICAL_VIOLATIONS = sum(CRITICAL.FLAG == "Critical", na.rm = TRUE),
    .groups = 'drop'
  ) %>%
  mutate(
    # Assign letter grade for each inspection
    GRADE = case_when(
      SCORE <= 13 ~ "A",
      SCORE <= 27 ~ "B",
      TRUE ~ "C"
    )
  ) %>%
  arrange(CAMIS, INSPECTION.DATE)

head(restaurant_unique)
```

```
## # A tibble: 6 × 6
##   CAMIS DBA                INSPECTION.DATE SCORE CRITICAL_VIOLATIONS GRADE
##   <int> <chr>                <date>         <int>         <int> <chr>
## 1 30075445 MORRIS PARK BAKE SHOP 2023-01-31         21             5 B
## 2 30075445 MORRIS PARK BAKE SHOP 2023-02-03         13             1 A
## 3 30075445 MORRIS PARK BAKE SHOP 2023-08-01         38             2 C
## 4 30075445 MORRIS PARK BAKE SHOP 2023-08-22         12             1 A
## 5 30075445 MORRIS PARK BAKE SHOP 2024-11-08         10             1 A
## 6 30191841 D.J. REYNOLDS      2023-04-23         10             2 A
```

Analyzing the Data

```
# For each restaurant, compare current vs previous inspection
restaurant_persistence <- restaurant_unique %>%
  group_by(CAMIS) %>%
  arrange(INSPECTION.DATE) %>%
  mutate(
    score_prev = lag(SCORE), # Previous inspection score
    score_change = SCORE - score_prev,
    critical_prev = lag(CRITICAL_VIOLATIONS),

    # Indicates if critical violation is recurring
    critical_recur = ifelse(CRITICAL_VIOLATIONS > 0 & critical_prev > 0, 1, 0)
  ) %>%
  filter(!is.na(score_prev)) %>% #Remove first inspections
  ungroup()

head(restaurant_persistence)
```

```
## # A tibble: 6 × 10
##   CAMIS DBA      INSPECTION.DATE SCORE CRITICAL_VIOLATIONS GRADE score_prev
##   <int> <chr>    <date>          <int>          <int> <chr>      <int>
## 1 40944945 STAGE 42    2015-10-14         5             0 A         2
## 2 50041177 FIRSTLIVE... 2015-10-15         2             0 A        38
## 3 50041177 FIRSTLIVE... 2016-01-21         6             0 A         2
## 4 41611709 EQUESTRIS    2016-03-26        13             2 A        21
## 5 41627984 JOUVAY NI... 2016-04-02         7             1 A        19
## 6 40944945 STAGE 42    2016-06-10        18             2 B         5
## # i 3 more variables: score_change <int>, critical_prev <int>,
## #   critical_recur <dbl>
```

```
# Categorizes each inspection pair on whether the restaurant 'Improved', 'Worsened', or
# 'No Change' based on change in score
score_summary <- restaurant_persistence %>%
  mutate(
    score_trend = case_when(
      score_change < 0 ~ "Improved",
      score_change > 0 ~ "Worsened",
      TRUE ~ "No Change"
    )
  ) %>%
  count(score_trend) %>%
  mutate(prop = n / sum(n)) # Proportion of each category

print(score_summary)
```

```
## # A tibble: 3 × 3
##   score_trend      n    prop
##   <chr>         <int> <dbl>
## 1 Improved     28961 0.518
## 2 No Change    2890 0.0517
## 3 Worsened    24047 0.430
```

```
## Computing proportions and count for recurrence of critical violations
critical_summary <- restaurant_persistence %>%
  summarise(
    total_pairs = n(),
    critical_recur_count = sum(critical_recur),
    prop_critical_recur = critical_recur_count / total_pairs
  )

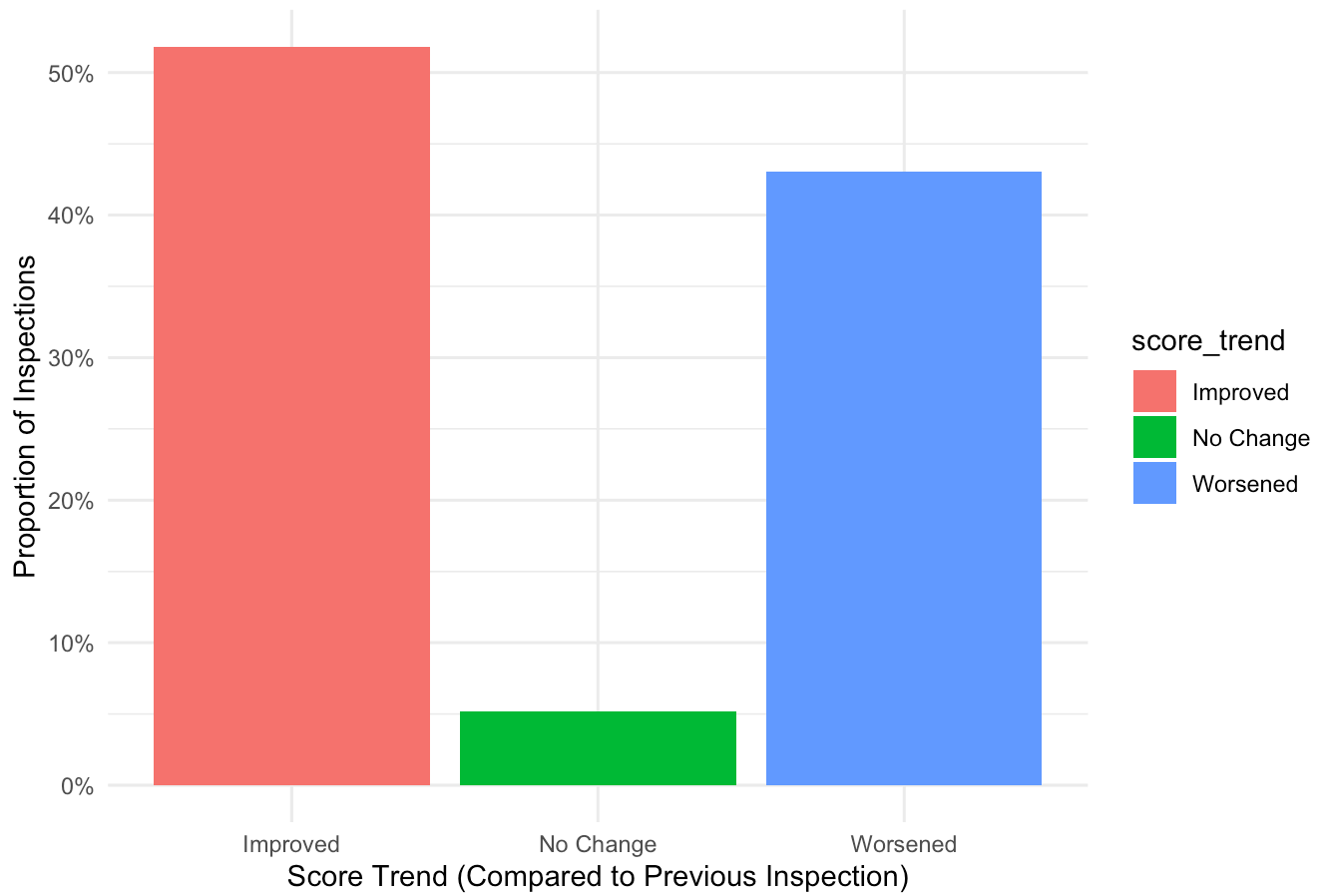
print(critical_summary)
```

```
## # A tibble: 1 × 3
##   total_pairs critical_recur_count prop_critical_recur
##   <int>         <dbl>         <dbl>
## 1      55898          43366          0.776
```

Data Visualizations

```
# Plot Score Trends (Improved / No Change / Worsened)
ggplot(score_summary, aes(x=score_trend, y=prop, fill=score_trend)) +
  geom_bar(stat = "identity") +
  scale_y_continuous(labels = percent_format()) +
  labs(title = "Improvement vs. Decline Across The Boroughs",
       x = "Score Trend (Compared to Previous Inspection)",
       y = "Proportion of Inspections") +
  theme_minimal()
```

Improvement vs. Decline Across The Boroughs



```
ggsave("score_trend_barplot.png", width = 6, height = 4, dpi = 300)
```

```

restaurant_yearly <- restaurant_persistence %>%
  mutate(year = format(INSPECTION.DATE, "%Y")) %>%
  group_by(year) %>%
  filter(year != "2015") %>%
  summarise(
    total_pairs = n(),
    critical_recur_count = sum(critical_recur),
    prop_critical_recur = critical_recur_count / total_pairs
  )

# Yearly Inspeciton Counts + Recurrence Rate on Log Scale
ggplot(restaurant_yearly, aes(x=year)) +
  geom_bar(
    aes(y=total_pairs),
    stat="identity",
    fill="lightblue",
    alpha=0.7
  ) +
  geom_line(
    aes(y=prop_critical_recur * 1000, color="Recurrence Proportion", group=1),
    size=1.2
  ) +
  geom_point(
    aes(y=prop_critical_recur * 1000, color="Recurrence Proportion"),
    size=2
  ) +
  scale_y_continuous(
    name="Number of Inspection Pairs",
    trans="log10",
    sec.axis = sec_axis(
      ~ . / 1000,
      labels=scales::percent_format(),
      name="Recurrence Proportion")
  ) +
  scale_color_manual(values= c("Recurrence Proportion"="darkred")) +
  labs(
    title="Log-Scaled Inspections Over Time",
    x="Year",
    color="Metric") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1)
  )

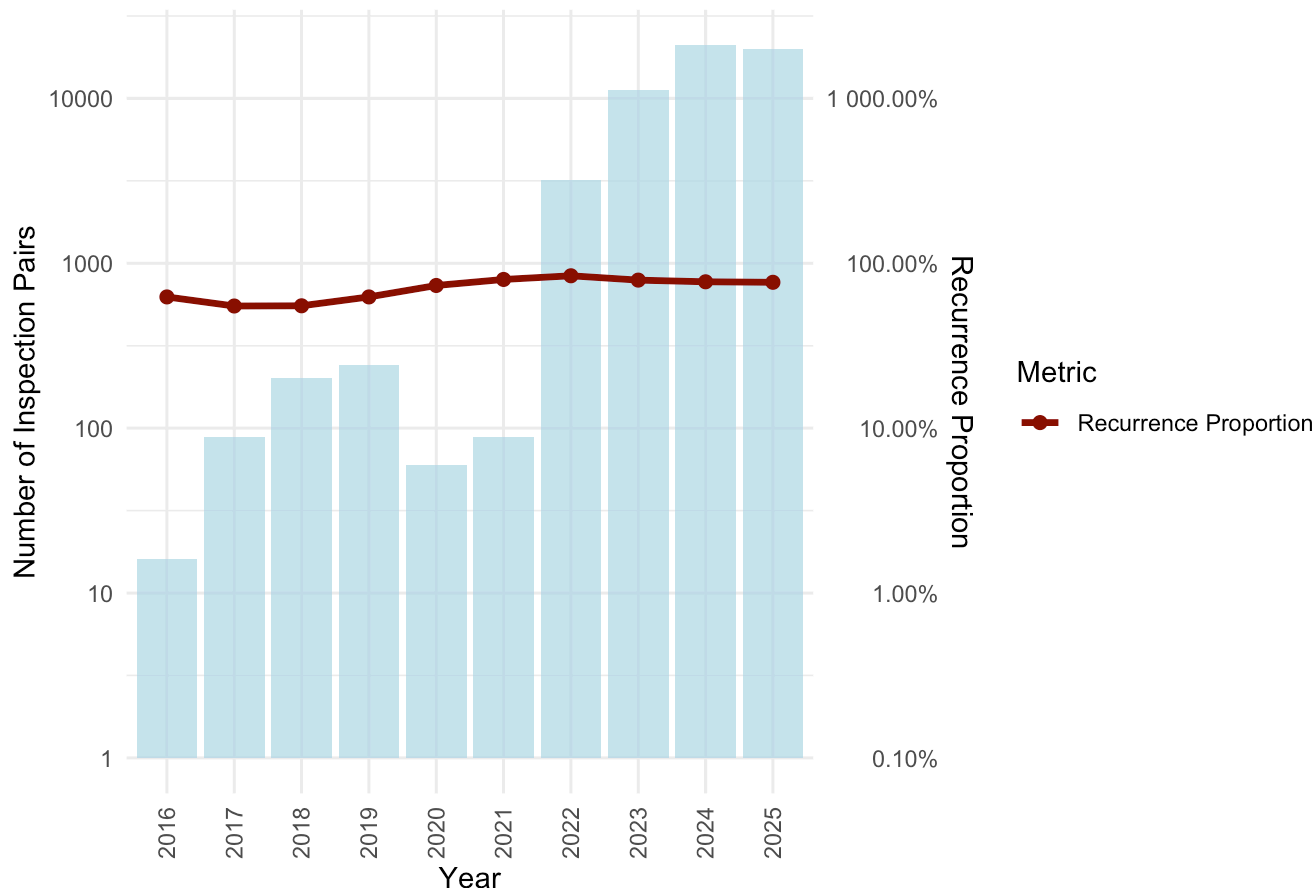
```

```

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

Log-Scaled Inspections Over Time



```
ggsave("yearly_recidivism_plot.png", width =8, height=5, dpi = 300)
```

```
# McNemar's Test for Recurrence of Critical Violations (AI-SUGGESTED IDEA)
```

```
# Contingency Table:
```

```
# Did a restaurant have critical violations previously vs currently?
```

```
tab <- table(
  prev = restaurant_persistence$critical_prev > 0,
  curr = restaurant_persistence$CRITICAL_VIOLATIONS > 0
)
```

```
mcnemar.test(tab)
```

```
##
```

```
## McNemar's Chi-squared test with continuity correction
```

```
##
```

```
## data: tab
```

```
## McNemar's chi-squared = 152.57, df = 1, p-value < 2.2e-16
```

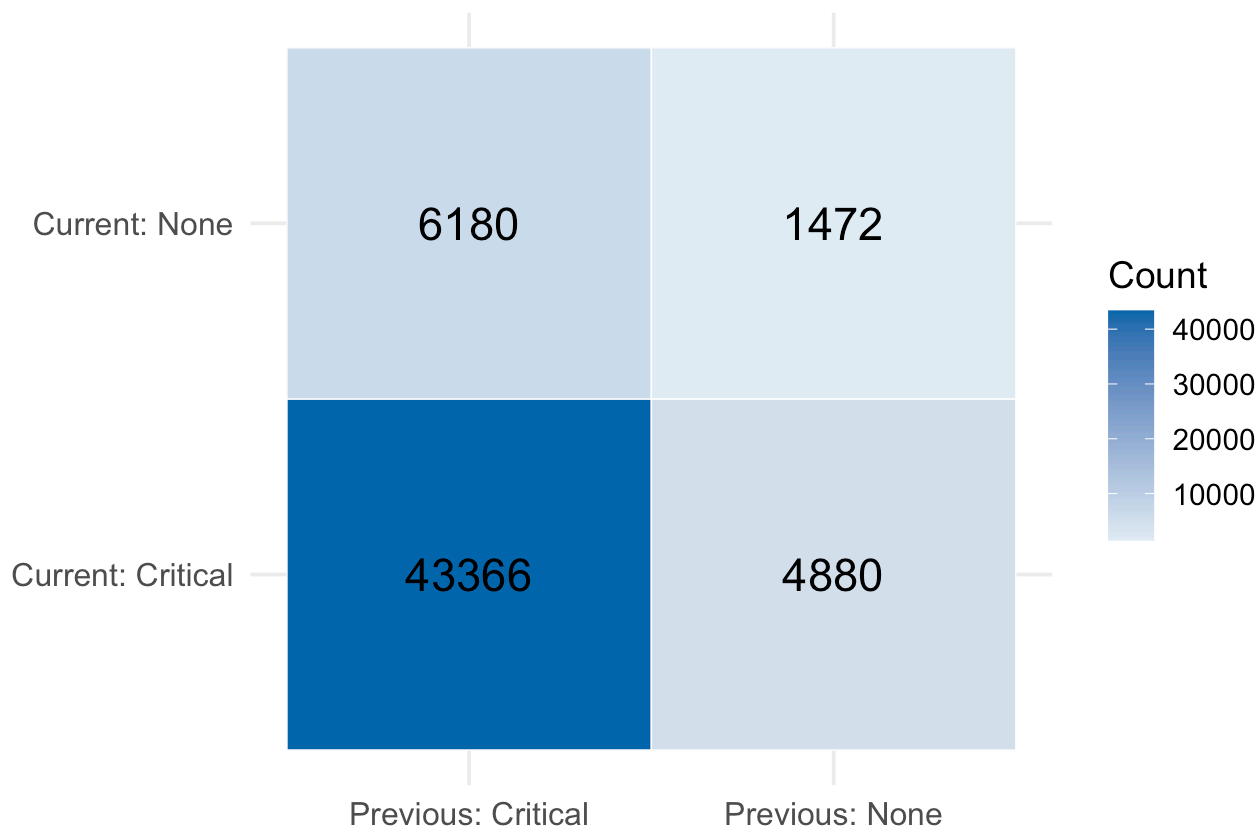
The McNemar's test gave a chi-squared statistic of 152.57 with 1 degree of freedom, as well as a p-value < $2.2e^{-16}$, which is far below any conventional significance level. These results indicate that restaurants with prior critical violations are significantly more likely to have violations in subsequent inspections, suggesting persistence rather than random occurrence.

Visualizing McNemar Contingency Table

```
#Converting the dataframe into a heatmap
mcnemar_df <- restaurant_persistence %>%
  mutate(
    prev = ifelse(critical_prev > 0, "Previous: Critical", "Previous: None"),
    curr = ifelse(CRITICAL_VIOLATIONS > 0, "Current: Critical", "Current: None")
  ) %>%
  count(prev, curr)

ggplot(mcnemar_df, aes(x = prev, y = curr, fill = n)) +
  geom_tile(color = "white") +
  geom_text(aes(label = n), size = 6, color = "black") +
  scale_fill_gradient(low = "#e0ecf4", high = "#0868ac") +
  labs(
    title = "Recurrence of Critical Violations",
    x = "",
    y = "",
    fill = "Count",
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "right",
    axis.text.x = element_text(size=12),
    axis.text.y = element_text(size=12)
  )
```

Recurrence of Critical Violations



```
ggsave("critical_violation_heatmap.png", width=6, height=5, dpi=300)
```

HIGH-RISK CUISINES AND BOROUGHES

```
restaurant <- read.csv("newyork_restaurant_inspections.csv")  
head(restaurant)
```

```

##          CAMIS                      DBA          BORO BUILDING
## 1 50179007          STOUT MARKETS INC Manhattan    1328
## 2 50167375          181Exclusive LLC Manhattan    810
## 3 50178660          PAPACHENCHARSIUBAO CORP  Queens  61-17
## 4 50111182          THE ALCHEMISTS KITCHEN Manhattan    117
## 5 50075461          POPEYES Manhattan    2300
## 6 50175948 A DAUGHTER AND TWO SON'S LLC  Bronx    1807
##          STREET ZIPCODE          PHONE CUISINE.DESCRPTION
## 1          BROADWAY    10001 7187919245
## 2          WEST 181 STREET    10033 2127950550
## 3          SPRINGFIELD BOULEVARD    11364 3472350424
## 4          CROSBY STREET    10012 9177433747
## 5 ADAM CLAYTON POWELL JR BOULEVARD    NA 6463689765          Chicken
## 6          ARCHER STREET    10460 3477204188
##  INSPECTION.DATE                      ACTION
## 1          01/01/1900
## 2          01/01/1900
## 3          01/01/1900
## 4          01/01/1900
## 5          04/30/2024 Violations were cited in the following area(s).
## 6          01/01/1900
##  VIOLATION.CODE          VIOLATION.DESCRPTION  CRITICAL.FLAG SCORE GRADE
## 1                                Not Applicable    NA
## 2                                Not Applicable    NA
## 3                                Not Applicable    NA
## 4                                Not Applicable    NA
## 5          09B Thawing procedure improper.    Not Critical    22
## 6                                Not Applicable    NA
##  GRADE.DATE RECORD.DATE                      INSPECTION.TYPE Latitude
## 1          11/24/2025                                40.75031
## 2          11/24/2025                                40.85103
## 3          11/24/2025                                40.74880
## 4          11/24/2025                                40.72426
## 5          11/24/2025 Cycle Inspection / Initial Inspection  0.00000
## 6          11/24/2025                                40.83754
##  Longitude Community.Board Council.District Census.Tract    BIN    BBL
## 1 -73.98738          105          4          7600 1015867 1008360001
## 2 -73.93876          112          10          26500 1064298 1021770100
## 3 -73.75655          411          23          137700 4163156 4076440009
## 4 -73.99662          102          1          4300 1007930 1005100006
## 5    0.00000          NA          NA          NA    NA    1
## 6 -73.86693          209          18          21800 2028633 2039200024
##  NTA                      Location
## 1 MN17 POINT (-73.987378651873 40.750308585404)
## 2 MN36 POINT (-73.938763711348 40.851030689063)
## 3 QN42 POINT (-73.756547103858 40.748799391976)
## 4 MN24 POINT (-73.9966159052 40.724261421332)
## 5
## 6 BX08 POINT (-73.866927323367 40.837537207937)

```

#cleaning for EDA

```

restaurant <- restaurant[, c("CAMIS","BORO", "CUISINE.DESCRPTION", "INSPECTION.DATE",
"SCORE","VIOLATION.CODE","VIOLATION.DESCRPTION","GRADE")]
restaurant$INSPECTION.DATE <- as.Date(restaurant$INSPECTION.DATE, format = "%m/%d/%Y")
restaurant$YEAR <- as.numeric(format(restaurant$INSPECTION.DATE, format = "%Y"))
restaurant <- restaurant[!is.na(restaurant$SCORE), ]
restaurant <- restaurant[restaurant$BORO != "" & restaurant$BORO != "0", ]
restaurant$CUISINE.DESCRPTION <-
toupper(restaurant$CUISINE.DESCRPTION)
restaurant <- restaurant[restaurant$SCORE <= 60, ]
head(restaurant)

```

```

##          CAMIS          BORO          CUISINE.DESCRPTION  INSPECTION.DATE  SCORE
## 5  50075461  Manhattan          CHICKEN      2024-04-30      22
## 13 41668282  Manhattan          ITALIAN      2025-07-17      23
## 21 50066463  Manhattan          AMERICAN      2022-04-18      13
## 26 50129114   Queens  BAKERY PRODUCTS/DESSERTS      2023-08-11      38
## 36 50049514   Queens          CREOLE      2021-09-03      39
## 42 40387418  Manhattan          BAGELS/PRETZELS      2025-07-14      26
##          VIOLATION.CODE
## 5          09B
## 13          04L
## 21          02B
## 26          04A
## 36          04L
## 42          04L
##
##                                     VIOLATION.DESCRPTION
## 5                                     Thawing procedure improper.
## 13          Evidence of mice or live mice in establishment's food or non-food areas.
## 21                                     Hot food item not held at or above 140° F.
## 26 Food Protection Certificate (FPC) not held by manager or supervisor of food operations.
## 36          Evidence of mice or live mice present in facility's food and/or non-food areas.
## 42          Evidence of mice or live mice in establishment's food or non-food areas.
##          GRADE  YEAR
## 5          2024
## 13          Z 2025
## 21          A 2022
## 26          2023
## 36          2021
## 42          Z 2025

```

#Summaries

##Most common cuisines

```
cuisine_count <- sort(table(restaurant$CUISINE.DESCRPTION), decreasing = TRUE)
top10_cuisine <-head(cuisine_count, 10)
```

American cuisine has the most occurrence

##Most inspected Boroughs

```
sort(table(restaurant$BORO), decreasing = TRUE)
```

```
##
##      Manhattan      Brooklyn      Queens      Bronx Staten Island
##           96419           66956           64294           24463           9352
```

Manhattan has the most occurrence ##Score Statistic

```
summary(restaurant$SCORE)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00   12.00   20.00   22.09   30.00   60.00
```

Visualization

```
library(ggplot2)
```

1. Top 10 Most Common Cuisines

```
barplot(top10_cuisine, main = "Top 10 Cuisine Types by Frequency", ylab = "Number of Ins
pections", color = "lightblue")
```

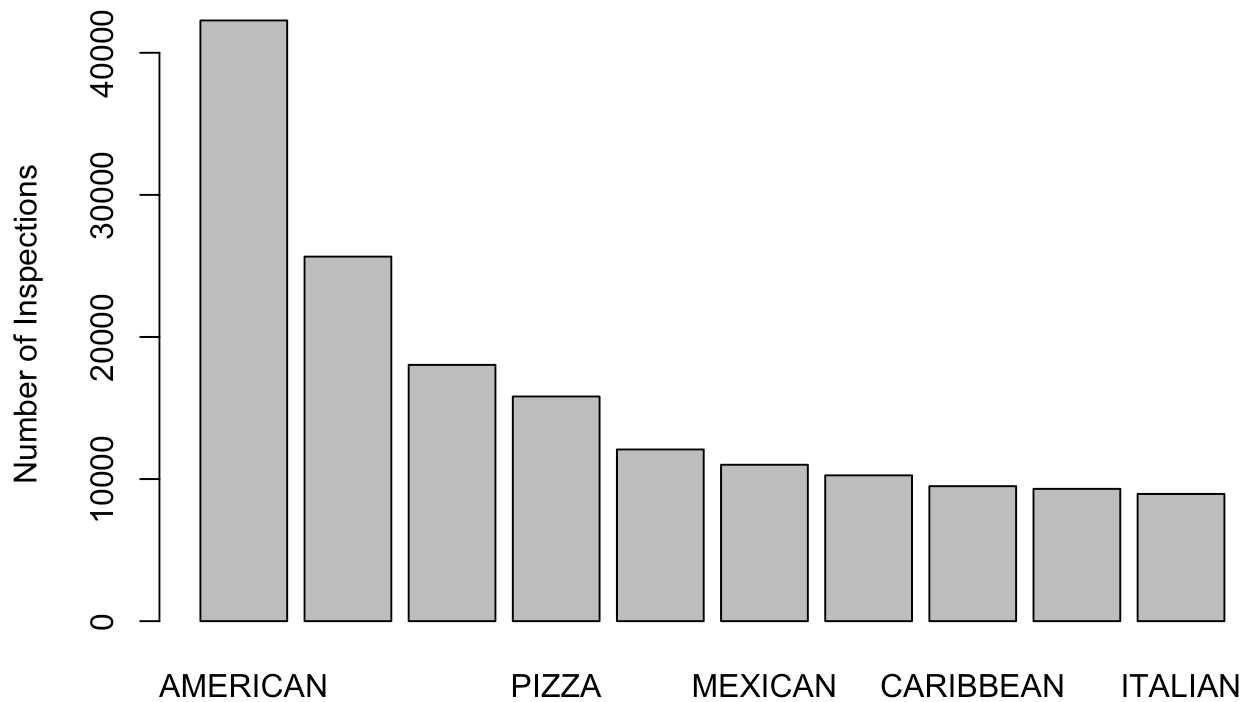
```
## Warning in plot.window(xlim, ylim, log = log, ...): "color" is not a graphical
## parameter
```

```
## Warning in axis(if (horiz) 2 else 1, at = at.l, labels = names.arg, lty =
## axis.lty, : "color" is not a graphical parameter
```

```
## Warning in title(main = main, sub = sub, xlab = xlab, ylab = ylab, ...):
## "color" is not a graphical parameter
```

```
## Warning in axis(if (horiz) 1 else 2, cex.axis = cex.axis, ...): "color" is not
## a graphical parameter
```

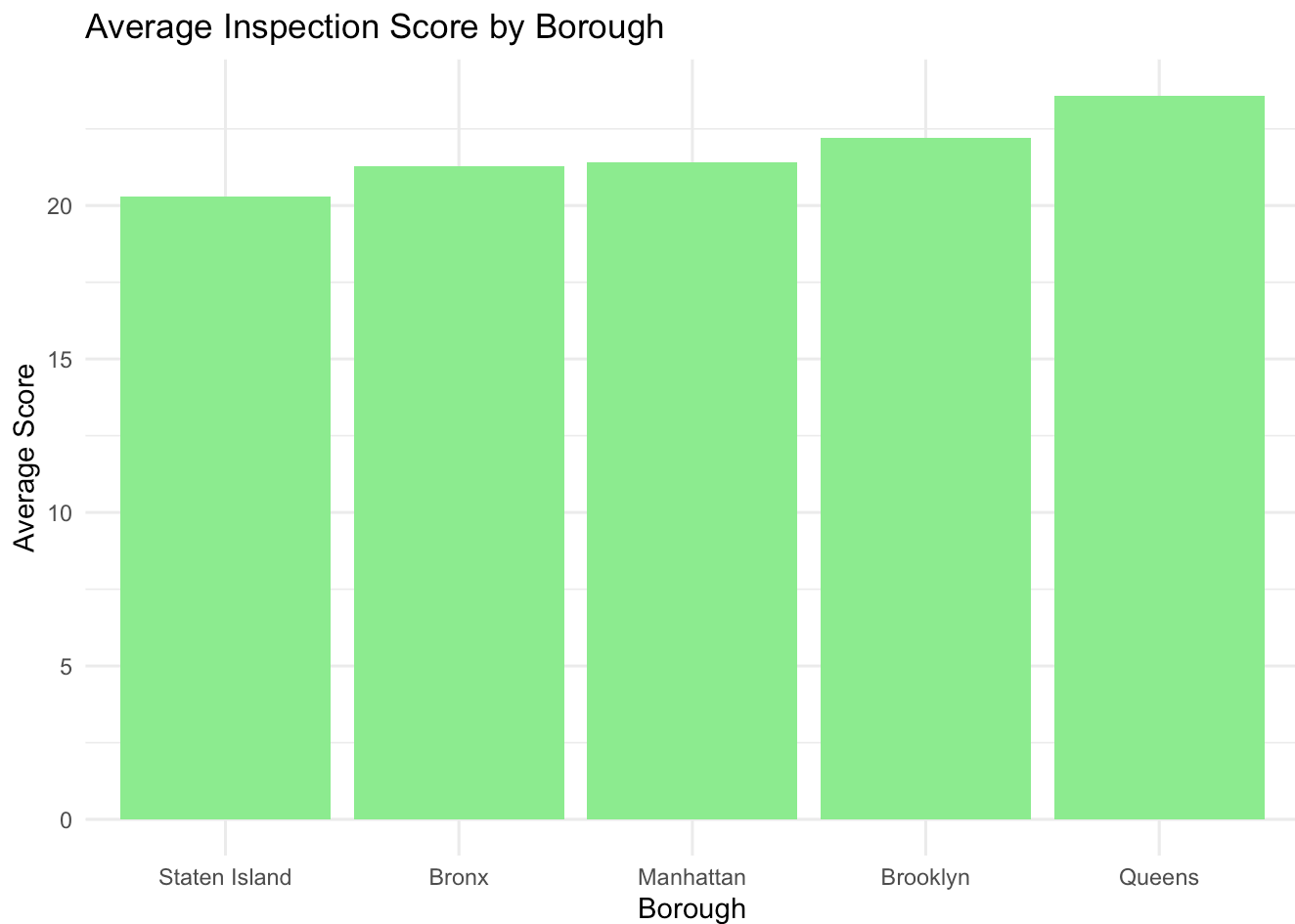
Top 10 Cuisine Types by Frequency



2. Average Inspection Score by Borough

```
avg_boro <- aggregate(SCORE ~ BOR0, data = restaurant, FUN = mean)

ggplot(avg_boro, aes(x = reorder(BOR0, SCORE), y = SCORE)) +
  geom_col(fill = "lightgreen") +
  labs(title = "Average Inspection Score by Borough", x = "Borough", y = "Average Score") +
  theme_minimal()
```



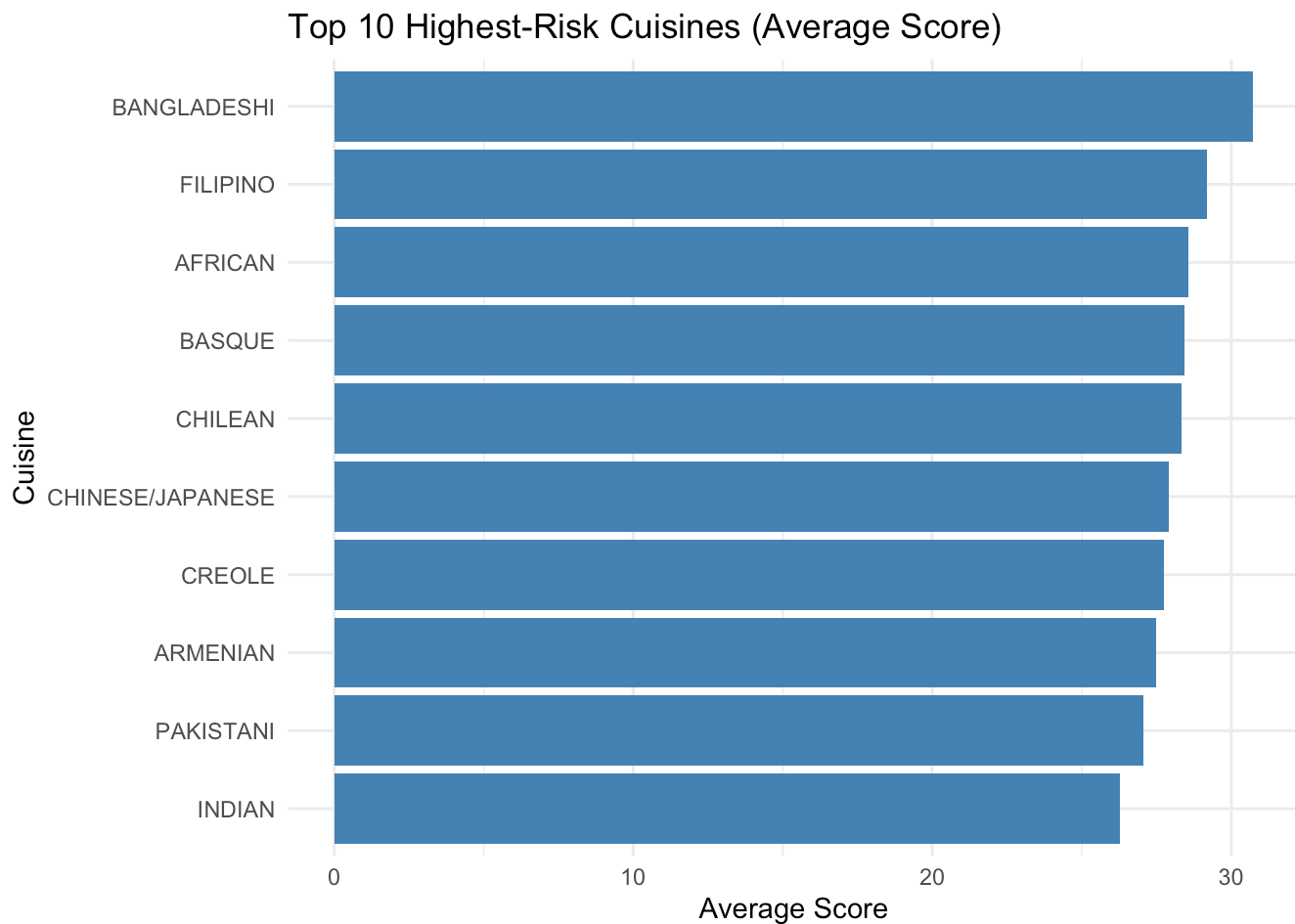
```
ggsave("avg_score_by_borough.png", width=8, height=6, dpi=300)
```

##3.Cuisines With the Highest Average Scores (Worst Overall Food Safety Performance)

```
avg_cuisine <- aggregate(SCORE ~ CUISINE.DESCRPTION, data = restaurant, mean)

worst10 <- avg_cuisine[order(avg_cuisine$SCORE, decreasing = TRUE),][1:10, ]

ggplot(worst10, aes(x = reorder(CUISINE.DESCRPTION, SCORE), y = SCORE)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "Top 10 Highest-Risk Cuisines (Average Score)",
       x = "Cuisine",
       y = "Average Score") +
  theme_minimal()
```



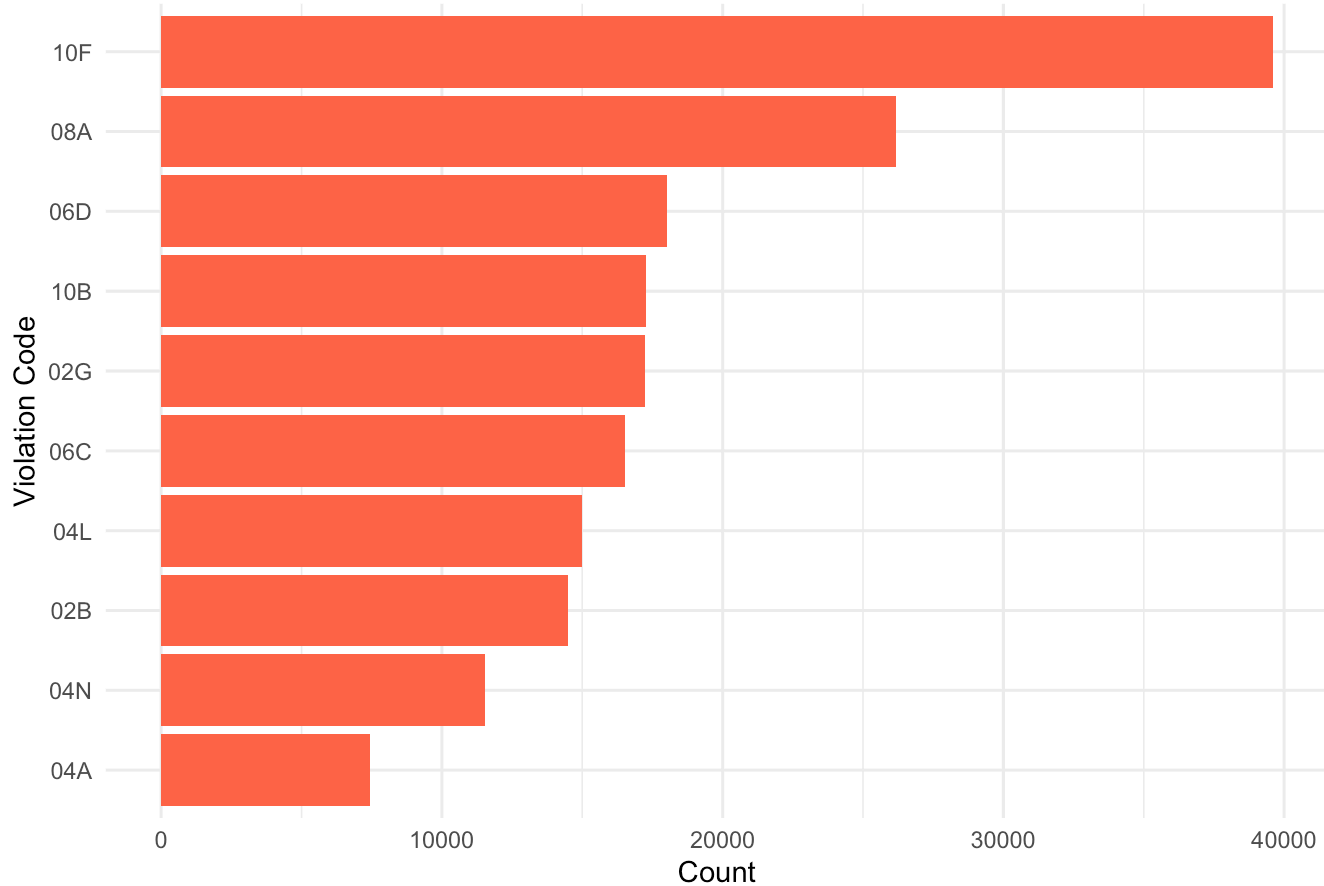
```
ggsave("high_risk_cuisines_avg.png", width=8, height=6, dpi=300)
```

##Top 10 Violation Codes

```
viol_counts <- sort(table(restaurant$VIOLATION.CODE), decreasing = TRUE)
top10_viol <- head(viol_counts, 10)
top10_viol_df <- data.frame( ViolationCode = names(top10_viol), Count= as.numeric(top10_viol), stringsAsFactors = FALSE)

ggplot(top10_viol_df, aes(x = reorder(ViolationCode, Count), y = Count)) +
  geom_col(fill = "tomato") +
  coord_flip() +
  labs(
    title = "Top 10 Most Frequent Violation Codes",
    x = "Violation Code",
    y = "Count"
  ) +
  theme_minimal()
```

Top 10 Most Frequent Violation Codes



```
ggsave("frequent_violation_codes.png", width=8, height=6, dpi=300)
```

10F: Non-food contact surface or equipment made of unacceptable material, not kept clean, or not properly sealed, raised, spaced or movable to allow accessibility for cleaning on all sides, above and underneath the unit. This is the most common violationnn ## prediction

```
# Create BAD_GRADE variable: 1 = B/C, 0 = A
restaurant$BAD_GRADE <- ifelse(restaurant$GRADE %in% c("B", "C"), 1, 0)

# Keep relevant columns
ml_data <- restaurant[, c("BAD_GRADE", "SCORE")]

# Remove missing values
ml_data <- na.omit(ml_data)

# Fit simple logistic regression
model <- glm(BAD_GRADE ~ SCORE,
             data = ml_data,
             family = binomial)

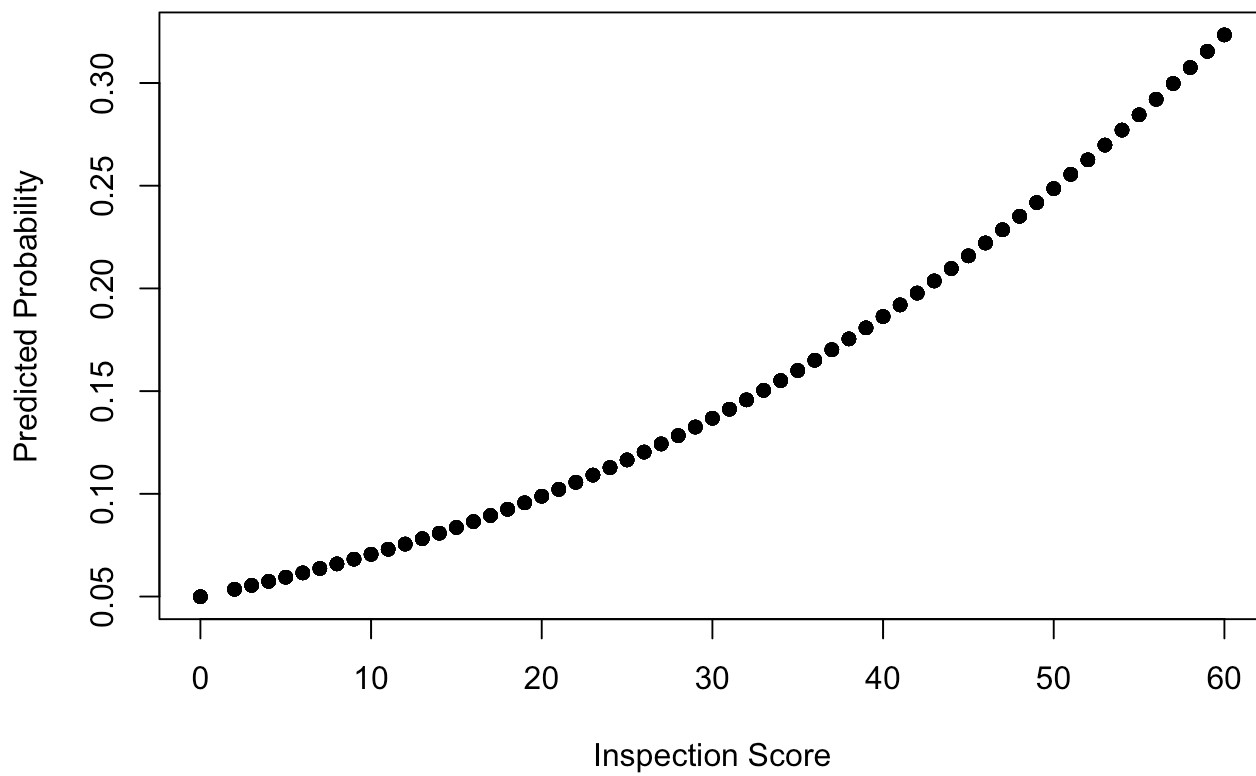
summary(model)
```

```
##
## Call:
## glm(formula = BAD_GRADE ~ SCORE, family = binomial, data = ml_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.9459479  0.0131207 -224.53  <2e-16 ***
## SCORE       0.0367969  0.0004198   87.65  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 187454  on 261483  degrees of freedom
## Residual deviance: 179891  on 261482  degrees of freedom
## AIC: 179895
##
## Number of Fisher Scoring iterations: 5
```

The logistic regression shows that as inspection scores increase (indicating more violations), the probability of receiving a bad grade (B or C) increases significantly.

```
plot(ml_data$SCORE, fitted(model),
     pch = 16, col = "#00000022",
     main = "Predicted Probability of Bad Grade vs. Score",
     xlab = "Inspection Score",
     ylab = "Predicted Probability")
```

Predicted Probability of Bad Grade vs. Score



Observation the logistic curve confirms a strong positive relationship between bad grades and violations: As inspection scores increase, the likelihood of receiving a bad grade increases sharply.